

EL CALAFATE INTL, Argentina

WMO: 879040

Lat: 50.2844S

Lon: 72.0514W

Elev: 669

StdP: 14.34

Time Zone: -3.00 (W03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	16.5	20.2	12.5	10.6	27.3	15.4	12.3	32.7	32.3	45.1	29.6	46.5	4.5	140	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.6	72.0	55.5	69.4	53.9	67.0	52.4	57.0	69.7	55.3	67.4	53.8	65.1	16.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.8	56.4	60.6	48.6	52.0	58.8	46.9	48.7	58.0	24.7	69.8	23.6	67.2	22.7	65.5	64.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.9	29.5	26.7	DB	11.3	77.8	5.1	3.4	7.6	80.3	4.6	82.3	1.7	84.2	-2.0	86.6	(4)
(5)			WB	10.8	60.1	5.1	1.8	7.2	61.4	4.2	62.4	1.4	63.5	-2.3	64.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	56.1	56.7	52.1	45.0	39.4	34.2	34.4	36.7	41.7	45.9	50.5	53.7
	DBStd	9.47	4.73	5.23	4.82	5.27	5.62	6.53	5.99	5.21	4.62	4.59	5.01	4.47
	HDD50	2374	7	5	31	165	330	476	485	411	251	142	55	16
	HDD65	7135	276	238	399	599	793	924	950	876	700	593	435	352
	CDD50	723	197	192	98	17	2	1	0	0	2	14	70	130
	CDD65	7	2	5	0	0	0	0	0	0	0	0	0	0
	CDH74	37	11	26	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.9	14.8	12.7	11.2	9.8	7.9	7.7	7.4	8.0	9.9	11.9	14.4	14.6
Precipitation	PrecAvg	7.2	0.4	0.4	0.7	1.0	1.1	0.9	0.8	0.7	0.6	0.4	0.3	0.6
	PrecMax	12.8	0.9	1.0	4.1	2.8	2.8	2.6	1.8	1.3	1.7	1.2	1.1	2.0
	PrecMin	3.1	0.1	0.0	0.0	0.1	0.2	0.2	0.2	0.3	0.1	0.0	0.0	0.0
	PrecStd	2.6	0.2	0.3	1.1	0.8	0.7	0.7	0.5	0.3	0.5	0.4	0.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	77.7	70.5	65.8	58.9	57.0	53.6	55.0	59.4	65.1	69.6	70.7
		MCWB	56.7	57.8	54.4	52.7	48.5	46.3	43.4	44.1	46.2	47.9	53.1	53.5
	2%	DB	71.2	73.4	67.1	60.9	55.0	50.8	49.5	51.2	56.1	60.9	65.5	68.0
		MCWB	54.8	57.2	52.7	49.1	46.0	42.8	41.3	42.2	44.3	46.7	50.5	52.2
	5%	DB	68.5	70.4	64.4	57.9	52.0	47.1	46.5	48.6	53.6	58.2	62.9	65.3
		MCWB	53.1	55.2	51.4	47.5	43.8	40.4	39.4	40.5	42.9	45.5	49.1	50.3
	10%	DB	66.1	67.2	61.9	55.4	49.2	44.1	44.1	46.0	51.2	56.0	60.5	62.8
		MCWB	52.0	53.5	50.4	46.3	42.1	38.4	37.9	39.1	41.6	44.3	47.8	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.2	56.1	54.2	50.3	46.5	44.8	45.6	47.3	49.4	54.5	56.5
		MCDB	70.2	72.2	67.3	64.4	57.1	55.8	52.3	52.9	56.5	62.0	66.4	66.1
	2%	WB	56.1	57.9	54.2	50.4	47.4	43.5	42.0	43.4	45.5	47.5	51.9	53.1
		MCDB	69.0	70.8	64.5	58.1	53.2	49.7	48.6	49.2	53.9	58.6	62.9	65.1
	5%	WB	54.5	56.0	52.7	48.7	45.0	41.0	40.2	41.6	44.0	46.2	50.1	51.3
		MCDB	66.6	68.7	62.3	56.3	51.0	46.6	45.8	47.4	51.3	56.8	60.9	63.0
	10%	WB	53.0	54.4	51.3	47.0	42.9	38.8	38.4	39.7	42.7	44.9	48.6	50.0
		MCDB	64.6	66.1	60.2	54.5	48.2	43.4	43.5	45.2	49.4	55.0	59.4	60.8
Mean Daily Temperature Range	5% DB	MDBR	17.0	17.6	17.2	17.9	15.0	13.0	13.9	14.9	17.3	18.3	17.0	16.1
		MCDBR	20.9	21.7	19.7	20.5	18.1	15.5	16.1	18.0	20.3	21.5	20.1	19.4
		MCWBR	10.8	11.6	11.2	12.9	11.8	10.7	10.8	11.5	11.7	11.5	10.6	10.1
	5% WB	MCDBR	20.1	20.5	18.7	19.3	16.6	14.7	15.6	17.0	17.3	19.8	19.2	18.1
		MCWBR	11.2	12.4	11.7	12.7	11.8	10.8	11.0	11.7	10.8	11.4	10.6	10.2
Clear-Sky Solar Irradiance	taub	0.306	0.304	0.291	0.280	0.259	0.245	0.248	0.266	0.286	0.297	0.298	0.307	
	taud	2.564	2.576	2.608	2.607	2.546	2.426	2.408	2.458	2.501	2.533	2.555	2.542	
	Ebn at Noon	319	310	297	272	246	229	243	270	292	308	320	321	
	Edh at Noon	32	29	25	20	16	15	17	22	27	30	32	33	
All-Sky Solar Radiation	RadAvg	1931	1619	1125	683	386	265	326	567	971	1413	1814	1973	
	RadStd	113	84	64	40	33	22	31	50	72	73	99	127	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page